

SPDC in nonlinear photonic crystals

The generation of entangled photon pairs in nonlinear optical media can be described using a quantum mechanical interaction Hamiltonian, forming a cornerstone of emerging quantum technologies such as quantum communication and computation. The aim of this project is to study the spectral and angular emission properties of entangled photon pairs generated via **spontaneous parametric down-conversion (SPDC)** in a nonlinear photonic crystal with quadratic (second-order) nonlinearity. Using appropriate theoretical and numerical tools, you will analyze how energy and momentum conservation (phase-matching conditions) and determine the wavelengths and emission angles of the correlated signal and idler photons. The project will involve calculating these emission characteristics for a given crystal structure and pump configuration, and exploring how the periodic modulation of the nonlinear medium influences the phase-matching and spectral properties. Additionally, you may examine the underlying quantum mechanical description by considering the interaction Hamiltonian of the system, and use it to infer key features of the generated photon pairs, such as the correlations underpinning their entanglement. Visualization of emission spectra and angular distributions is encouraged to illustrate the connection between the crystal design and the resulting quantum optical behavior.

Spontaneous parametric down-conversion

Spontaneous parametric down-conversion (SPDC) is a nonlinear optical process in which a single high-energy **pump photon** (frequency ω_p) propagating in a nonlinear medium is converted into two lower-energy photons:

- **signal photon** (ω_s)
- **idler photon** (ω_i)

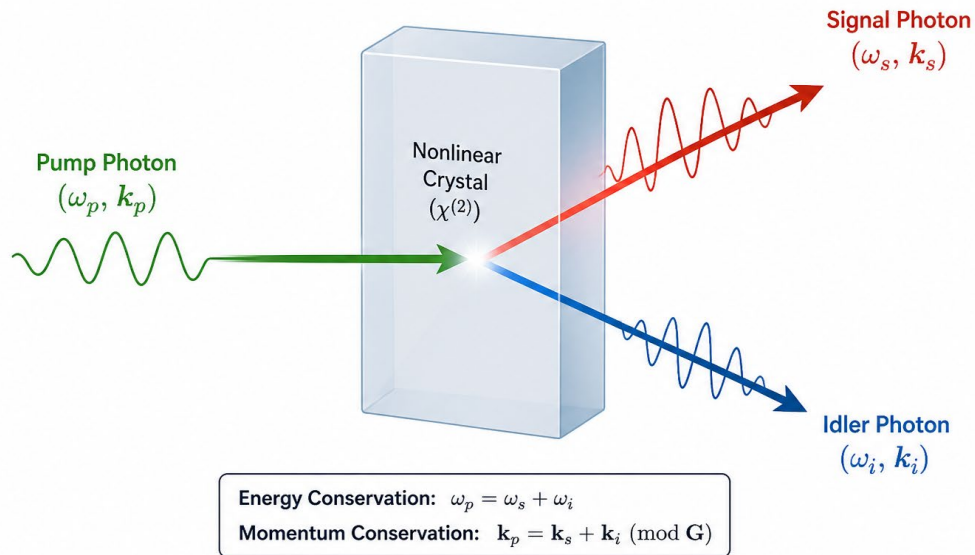
This process occurs in materials with a nonzero second-order optical susceptibility $\chi^{(2)}$, i.e. a **quadratic optical nonlinearity**. Key constraints in this three-particle interaction are:

1. **Energy conservation:** $\hbar\omega_p = \hbar\omega_s + \hbar\omega_i \Rightarrow \omega_p = \omega_s + \omega_i$
2. **Momentum conservation:** $\hbar\mathbf{k}_p = \hbar\mathbf{k}_s + \hbar\mathbf{k}_i \Rightarrow \mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i$

Condition number (2) is also known as the **phase matching condition** for the three-wave mixing process and links the propagation directions of the three photons, represented by the wavevectors $\mathbf{k}_m$ (with $m = p, s$ and i , for pump, signal and idler, respectively). Conditions (1) and (2) together determine the spectrum and emission angles of the signal and idler photons generated by SPDC (assuming a monochromatic pump).

Three-Photon Interaction: SPDC

$$\omega_p \rightarrow \omega_s + \omega_i$$



Correlated photon emission

In SPDC, the signal and idler photons are not created independently: they emerge from the same quantum event, consisting in the annihilation of a pump photon at ω_p (annihilation operator $\hat{a}_p$) and corresponding creation of a photon pair consisting of the (twin) photons at ω_s and at ω_i (creation operator $\hat{a}_s^\dagger \hat{a}_i^\dagger$). The quantum nature of the process leads to strong correlations, e.g.:

- If you measure ω_s you immediately know ω_i ($\omega_i = \omega_p - \omega_s$)
- The directions of propagation of the twin photons are also correlated by momentum conservation ($\mathbf{k}_i = \mathbf{k}_p - \mathbf{k}_s$)
- The signal and idler photons are also time-correlated as they are emitted essentially simultaneously

Note: In quantum mechanics (Dirac notation) the quantum state of the emitted photon pair is expressed as: $|\psi\rangle = \int d\omega_s d\omega_i f(\omega_s, \omega_i) |\omega_s\rangle_s |\omega_i\rangle_i$, where the function $f(\omega_s, \omega_i)$, known as the **joint spectral amplitude** (JSA), encodes the photon correlations (see also extra section at the end).

(Quasi) phase matching

In real materials, the refractive index varies with frequency, i.e.: $n = n(\omega)$. The frequency-dependent wave propagation constant ($k = |\mathbf{k}|$) is given by: $k(\omega) = \frac{\omega}{c} n(\omega)$, where ω is the (angular) frequency.

Momentum conservation (condition nr 2) then does not automatically follow from energy conservation (condition nr. 1). Efficient SPDC requires some engineering of the material properties

to bring the **phase mismatch** $\Delta\mathbf{k} = \mathbf{k}_p - \mathbf{k}_s - \mathbf{k}_i$ (close) to zero, i.e. $\Delta\mathbf{k} \approx 0$, otherwise the twin photon emission is suppressed by destructive interference. This problem can be solved by a technique known by **quasi phase-matching** (QPM), which can be realized with nonlinear photonic crystals.

Nonlinear photonic crystals

In your project, the nonlinear medium is a **nonlinear photonic crystal**, meaning that the second-order nonlinearity $\chi^{(2)}(\mathbf{r})$ is periodically modulated in space ($\mathbf{r}$). This nonlinear lattice introduces a **reciprocal lattice vector** $\mathbf{G}$, which contributes to the momentum conservation (phase matching) condition: $\mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i + \mathbf{G}$, resulting in a corrected effective phase mismatch given by:

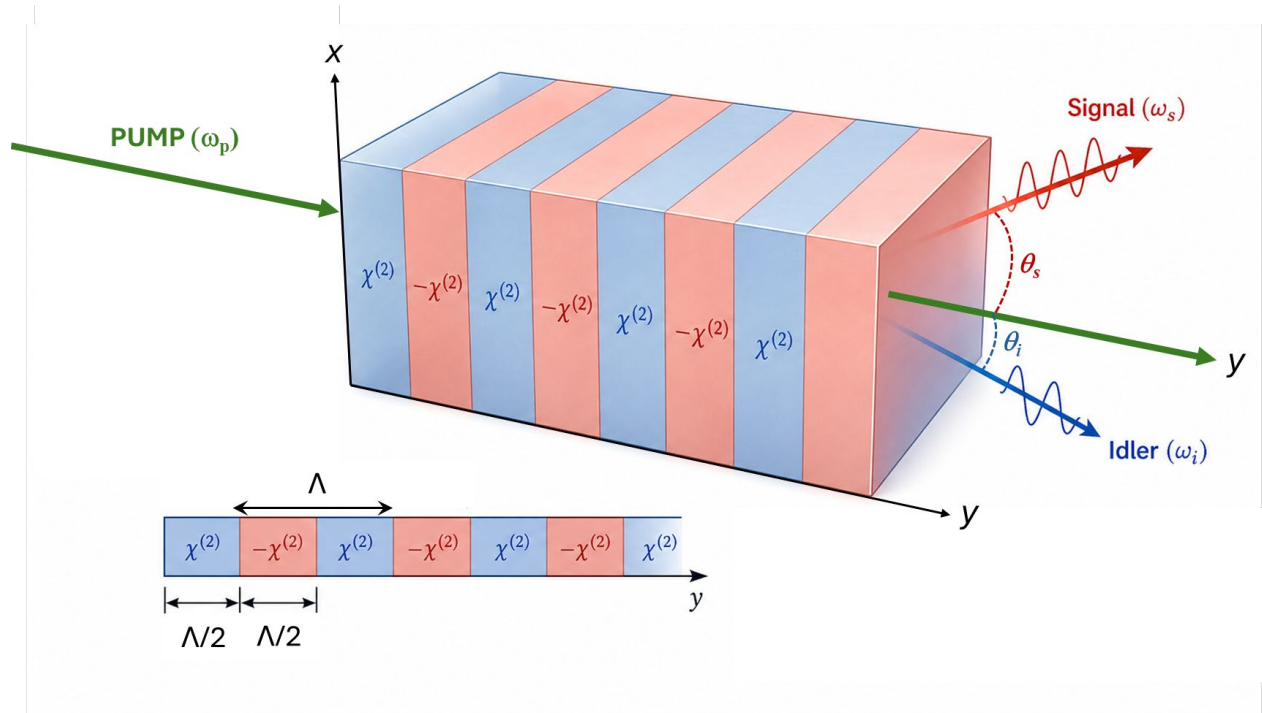
$\Delta\mathbf{k}_G = \mathbf{k}_p - \mathbf{k}_s - \mathbf{k}_i - \mathbf{G}$, which can be brought to zero by suitable choice of the spatial periodicity, so to compensate the intrinsic phase mismatch originating from the material dispersion ($\Delta\mathbf{k}$). In essence one can engineer phase matching by choosing the lattice periodicity and control the spectral and angular emission through suitable design of the nonlinear photonic crystal structure. Typical nonlinear photonic crystals are realized in ferroelectric material such as lithium niobate and lithium tantalate by a technology known as periodic poling, which spatially modulates nonlinearity between the values of $+\chi^{(2)}$ and $-\chi^{(2)}$.

Project tasks

Consider SPDC in a nonlinear crystal (1mol% MgO-doped nearly stoichiometric lithium tantalate in the experiments) with the following dispersion properties (Sellmeier equation for the refractive index n as a function of wavelength λ , specified in micrometers, and temperature T in degrees Celsius, with A - F coefficients given in the table below):

$$n^2(\lambda) = A + \frac{B + b(T)}{\lambda^2 - [C + c(T)]^2} + \frac{E}{\lambda^2 - F^2} + D\lambda^2$$

The crystal nonlinearity is modulated between the values of $+\chi^{(2)}$ and $-\chi^{(2)}$ in a square-wave fashion along the y -axis of the crystal, by a periodically poled 1D grating with 50% duty cycle and constant period $\Lambda = 8.5\mu\text{m}$. The crystal is kept at a temperature $T = 80^\circ$ and pumped by a green laser at a wavelength $\lambda_p = 532\text{ nm}$, propagating along the y -axis, as shown in the figure here below:



Considering QPM with the lowest-order reciprocal lattice vector of the 1D nonlinear grating, determine the signal and idler output wavelengths (λ_s and λ_i) and emission angles in the (x, y) plane, i.e. θ_s and θ_i , measured with respect to the pump propagation axis y .

Parameter	Description and value
T	Temperature in degrees Celsius
λ	Wavelength in μm
A	4.54773
B	0.0774167
C	0.22025
D	-0.0226143
E	2.39494
F	7.45352
$b(T)$	$4.23526 \times 10^{-8} \times (T + 273.15)^2$
$c(T)$	$-6.53227 \times 10^{-8} \times (T + 273.15)^2$

Illustrate the results by plotting the emission angle as a function of wavelength for the signal and idler photons. N.B.: these can be two separate plots showing the QPM curves (θ_s, λ_s) and (θ_i, λ_i) ,

Identify on the two diagrams the working points corresponding to twin photons which are both emitted collinearly with the pump (i.e. propagate along y). Determine the signal and idler wavelengths in this case, as well as those in which the signal is emitted at an angle of 1° (inside the crystal). What is the angle for the twin idler photon emission in the latter case?

Discussion

Report on the literature search, references and books used to address the assigned questions. Provide explanations for the models used and the procedures followed in your calculations. Plot and discuss the results, highlighting their key features.

Extra task

In the course we considered the time-independent Schrödinger equation for stationary states, separating the spatial and temporal variables. The corresponding stationary state for a photon is an harmonic oscillator, corresponding to the electric field at each single frequency. The SPDC process is then treated as a perturbation to this state of (uncoupled) harmonic oscillators at the three frequencies. Using (first-order) perturbation theory and a semiclassical approximation, where the pump field is not quantized and is not depleted (i.e. it is considered as a classical field, not a quantum operator, and described by a constant complex amplitude), the evolution of the system is described by the interaction Hamiltonian picture outlined here below. Using the Heisenberg equations of motion, one can then derive the expression describing the average photon number of the output signal and idler as a function of the pump photon rate (photon number per second) and the effective nonlinearity of the crystal, as well as highlight the special correlations (expressed by the second order correlation function) manifesting themselves in the signal and idler coincidence counts.

SPDC interaction Hamiltonian and equations of motion

In a $\chi^{(2)}$ medium, the total Hamiltonian of the system is written as:

$$\hat{H} = \hat{H}_0 + \hat{H}_I$$

where the unperturbed Hamiltonian is the sum of three oscillators (corresponding to the quantized electromagnetic waves) at the pump, signal and idler frequencies:

$$\hat{H}_0 = \sum_{m=p,s,i} \hbar \omega_m \left(\hat{N}_m + \frac{1}{2} \right)$$

where $\hat{N}_m = \hat{a}_m^\dagger \hat{a}_m$ is the photon number operator at frequency ω_m ($\hat{a}_m^\dagger$ and $\hat{a}_m$ being the photon creation and annihilation operators at the same frequency).

The SPDC interaction Hamiltonian takes instead the form (in the rotating wave approximation), assuming energy and momentum conservation to be satisfied:

$$\hat{H}_I = \hbar \kappa \left(\hat{a}_p \hat{a}_s^\dagger \hat{a}_i^\dagger + \hat{a}_p^\dagger \hat{a}_s \hat{a}_i \right)$$

where the coupling coefficient κ is a scalar parameter that incorporates the effect of the material effective nonlinearity and device length, ultimately expressing the medium's nonlinear efficiency (for perfect phase matching or QPM).

Using a semiclassical approximation, the (more intense) pump field for SPDC is treated as a classical undepleted field: The operator $\hat{a}_p$ in the interaction Hamiltonian is accordingly replaced by a constant (complex) field amplitude A_p , yielding the following interaction Hamiltonian:

$$\hat{H}_I = \hbar\kappa (A_p \hat{a}_s^\dagger \hat{a}_i^\dagger + A_p^* \hat{a}_s \hat{a}_i)$$

The average photon number rate (photon number per second) of the output signal and idler states $\langle \psi | \hat{N}_m | \psi \rangle$, with $m = s, i$ are then obtained as a function of the pump photon rate (related to A_p) and nonlinear efficiency of the crystal (related to κ).

Using (first order) time-dependent perturbation theory (Schrödinger picture) the output state can be obtained from the initial vacuum state as: $|\psi\rangle \approx |0\rangle + \frac{1}{i\hbar} \int dt H_I(t) |0\rangle$, but a more convenient approach is to move to the Heisenberg picture above, considering the time evolution of the operator rather than the Schrödinger picture (providing the time evolution of the state, expressed by the integral form). The dynamics of the SPDC process is then obtained directly by solving the Heisenberg equations of motion of the signal and idler operators, given by:

$$\frac{d\hat{O}}{dt} = -\frac{i}{\hbar} [\hat{O}, \hat{H}_I], \text{ with } \hat{O} = \hat{a}_s \text{ and } \hat{a}_i$$

Photon emission and coincidence rates

With the above formalism, determine:

1. the average signal and idler photon numbers, obtained as: $\langle \hat{N}_s \rangle = \langle \psi | \hat{a}_s^\dagger \hat{a}_s | \psi \rangle$ and $\langle \hat{N}_i \rangle = \langle \psi | \hat{a}_i^\dagger \hat{a}_i | \psi \rangle$
2. the signal and idler coincidence detection probability (associated to the second-order correlation function), given by: $\langle \psi | \hat{a}_s^\dagger \hat{a}_i^\dagger \hat{a}_s \hat{a}_i | \psi \rangle$

Note that in the Heisenberg notation (evolving operators) the solutions are found as e.g.:

$$\langle \hat{N}_s \rangle = \langle 0 | \hat{a}_s^\dagger(t) \hat{a}_s(t) | 0 \rangle.$$

If interested in further reading, see books by e.g. Mandel and Wolf, "*Optical coherence and quantum optics*", or Grynberg, Aspect and Fabre, "*Introduction to quantum optics*"